

Nizar Esmat Fawzy

Software Engineer
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PROFESSIONAL SUMMARY

Backend-focused Software Engineer building production systems in TypeScript, Node.js, and NestJS. Delivers 50+ REST API endpoints across 5 concurrent client platforms within a cross-functional team of 10+ engineers, owning API design, PostgreSQL schema architecture, and third-party integrations end-to-end. Also experienced with Python/Django, MongoDB, real-time systems (Socket.io), and testing with Jest.

TECHNICAL SKILLS

Programming Languages: TypeScript, JavaScript (ES6+), Python, SQL, C++
Backend: Node.js, NestJS, Express.js, RESTful API Design, OOP, Django
Real-Time & Messaging: Socket.io, WebSockets, Event-Driven Architecture
Databases: PostgreSQL, MongoDB, Schema Design, Data Modeling, Indexing, Query Optimization
Testing: Jest, Unit Testing, Integration Testing, Postman
Security & Auth: JWT, Authentication, Authorization, CORS, XSS/CSRF Mitigation, Input Validation
Tools & DevOps: Git, GitHub, Docker, Linux, Bash, CI/CD, Agile, SDLC
Frontend: React, Zustand

PROJECTS

- ECPC Movement Control** | *NestJS, PostgreSQL, TypeORM, Socket.io, React, JWT* GitHub
- Built a real-time volunteer dispatch system for the Egyptian Collegiate Programming Contest, replacing radio coordination with an auto-assignment engine that ranks volunteers by hall proximity, gender match, and idle time.
 - Modeled a TypeORM/PostgreSQL schema with versioned migrations and an append-only audit trail covering the full request lifecycle; streamed live updates to leader dashboards over Socket.io behind role-based JWT access.
- Chat Application** | *React, Node.js, Express.js, Socket.io, MongoDB, Cloudinary* GitHub Postman
- Engineered a full-stack real-time chat app (React/Zustand, Node/Express/Socket.io) with private messaging, group channels, live presence, and live message edit/delete sync.
 - Hardened the API with HTTP-only cookie JWT auth and Arcjet rate limiting against spam/abuse; integrated Cloudinary for image/voice storage with full CRUD.
- ITI Social Media Platform** | *PostgreSQL, REST API, React* GitHub
- Co-led backend architecture: designed normalized PostgreSQL schema, built RESTful APIs, and integrated with React.
 - Drove the full SDLC: requirements analysis, database modeling, backend development, and cross-team integration.

PROFESSIONAL EXPERIENCE

- Opream** May 2025 – Present
Software Engineer Cairo, Egypt
- Build and own 50+ RESTful API endpoints across 5 concurrent client platforms using TypeScript, NestJS, PostgreSQL, and React, collaborating within a cross-functional team of 10+ engineers.
 - Mostaqbel Watan Platform** – private social platform serving 1,000+ active users, with Jitsi video meetings, Socket.io private/group chat, and Facebook Graph API auto-ingestion of group posts.
 - Onway** – ride-share platform connecting riders and drivers, integrated with the Fawry payment gateway for secure transaction processing.
 - Al-Azhar & Bingo** – unified platform managing online and offline student lessons, plus a library management system.
 - Event Ticketing Platform** – implemented a scheduled job to auto-release stale “pending” ticket reservations back to “new” every 10 minutes, preventing inventory lock-up.
 - Improved database performance by normalizing schemas, adding targeted indexes, and eliminating N+1 query patterns; covered critical paths with Jest unit and integration tests.
- Information Technology Institute (ITI)** Nov 2024 – Mar 2025
Full Stack Developer Intern (Python) Smart Village, Egypt
- Designed a normalized PostgreSQL schema supporting 10k+ users with data integrity, indexing, and query optimization.
 - Delivered RESTful APIs with ETL-style data workflows and contributed to key architecture decisions.
- iSchool & WE School** Apr 2024 – Nov 2024
Software & Web Development Instructor (Part-time) Cairo, Egypt
- Mentored students in programming fundamentals and web development (HTML, CSS, JavaScript), building their software problem-solving skills.

EDUCATION

- Nahda University** Sep 2019 – Jul 2023
Bachelor of Computer Science Beni-Suef, Egypt
- Graduation Project: Smart Learning Management System with a Machine Learning model.

CERTIFICATIONS & ACTIVITIES

Languages: Arabic (Native), English (Fluent) | ECPC Contestant 2021 & 2022, 2nd Place | ICPC/ECPC/ACPC Volunteer, 2022–2024
Certifications: Python Professional Track (ITI) | SQL & PostgreSQL Developer’s Guide (Udemy) | Node.js Complete Guide (Udemy) | Full Stack MERN (Route Academy)